4 Information and Communication Technologies in Governance and Policies

Introduction

- Electronic governance, also referred to as e-governance, refers to the reformation of government operations through the use of information and communications technology (ICT).
- It is not the same as e-government because governance has a broader scope than government. e-governance can mean the way a country is run, the way an institution is run, or the way a housewife runs her own home. In this particular case, e-governance corresponds to the governance of a country.

e-Governance

- 'Governance' encompasses a broader scope than 'government.' Governance may be the activity of governing a country by its government, controlling a professional body by its executives, or controlling a household by the head of the household. Subsequently, e-governance may involve governing a country, organisation, company, or household, but with the assistance of ICT.
- However, in popular usage, e-governance refers to the use of ICT to govern a country/state. Thus, e-governance refers to the use of ICT to improve the efficiency, effectiveness, transparency, and accountability of information and transaction exchange.
  - Between governments
  - Between government agencies
  - Between government and citizens
  - Between government and businesses
- E-governance attempts to help individuals by providing them with information.

Types of Transaction in e-Governance

Governance is all about information flow between the government and citizens, the government and businesses, and within the government itself. All of these interactions are also covered by e-governance

- Government to citizen (G2C)
- Citizen to government (C2G)
- Government to government (G2G)
- Government to business (G2B)

Government to Citizen

- The most fundamental feature of e-governance is the link between the government and the citizen. In modern times, the government is involved in many facets of a citizen's life. The relationship between a citizen and the government begins with birth and ends with death. Every aspect of a person's life involves dealings with the government. It could be birth registration, marriage registration, divorce registration, or death registration.
- The G2C relationship will involve government services delivered to citizens. These services include telecommunications, transportation, postal services, and medical facilities.
- Electricity, education, and some democratic services related to citizenship such as certification, registration, licensing, taxation, passports, identification cards, and so on.
- As a result, e-governance in the G2C relationship will entail the use of ICT to facilitate the flow of services from the government to citizens (ICT).
- e-citizenship: e-citizenship will entail the use of ICT to facilitate government services associated with an individual's citizenship. It may include online transactions for the issuance and renewal of papers such as ration cards, passports, election cards, identity cards, and so on. It will be necessary for the government to construct a virtual identity for each citizen for them to access government services online. For the same purpose, the government needs to build a citizen database, which would be a massive undertaking.
- e-Registration: The term 'e-registration' refers to the online registration of certain contracts. Throughout a lifetime, a person

enters into multiple contracts. Most of these agreements must be registered to be lawful and enforceable. This type of registration can also be made ICT enabled. e-registration will assist to cut down on paperwork significantly.

- e-Transportation: e-transportation services comprise government services related to transportation by road, rail, water, or air. This includes
  - Booking and cancellation of tickets
  - Status of vehicles, railways, boats, and flights
  - Issue and renewal of driving licences
  - Registration and renewal of vehicles
  - Transfer of vehicles
  - Payment of the fees of licenses
  - Payment of fees and taxes for vehicle registration
- e-Health: e-health services is the ICT enablement of the government's health services. A linkage of all hospitals is possible. It is possible to establish a patient database. It is also possible to establish a local pharmacy database.
- e-Education: e-education encompass the use of ICT in the delivery of education and the delivery of courses. The usage of ICT will facilitate both remote and classroom education. The usage of the Internet can lessen the communication time necessary in distance education; it may also aid in the delivery of online programmes.
- e-Help: e-help is the use of ICT to facilitate disaster and crisis management. It entails the use of technology such as the Internet, SMS, and so on to decrease the response time of government agencies to disasters. Non-governmental organisations (NGOs) assist the government in providing disaster relief. Online information on disasters, warnings, and appeals for aid can allow the government and NGOs to organise their efforts and expedite rescue efforts.
- e-Taxation: By incorporating ICT into the taxing process, e-taxation will make the taxing procedure easier. Online tax due warnings and payment would facilitate transactions.

Citizen to Government (C2G)

- The C2G interaction will comprise citizen-to-government communication coming from the democratic process, such as voting, campaigning, feedback, and so on.
- Democracy: The genuine definition of democracy is public participation in the democratic processes. Citizens' active participation in the governance of the country is no longer possible due to the increased population. ICT can assist in enabling actual democratic processes such as voting, popular sentiment, criticism, and government accountability.
- e-Feedback: This is the use of ICT to provide input to the government. Lobbying is the process of persuading the government to make a specific decision. The usage of ICT can allow for inputs to the government as well as online debates about government services.

Government to Government (G2G)

- G2G ties include those between the central and state governments, as well as those between two or more government ministries.
- e-Administration: e-administration would entail the use of ICT in the administration's internal and external operations. ICT implementation can shorten communication time between government agencies. When utilised correctly, it can significantly minimise paperwork. e-Administration will help improve the morale and transparency of government administration.
- e-Police: The notion of e-police differs slightly from that of cyberpolice. To combat electronic/cybercrime, cyberpolice needs technological professionals. e-Police is the use of ICT to help the police department with investigation and administration. e-Police provides records of police officers, one's performance, criminal databases—wanted and in custody, crime patterns, and much more. ICT can assist the police department to save response time and expenses by eliminating paperwork.
- e-Courts: The concept of e-court incorporates the use of ICT to facilitate the judicial procedure.

Distance hearings, online summoning and orders, and online publication of verdicts and decrees may be aided by technology.

Government to Business (G2B)

- e-Taxation: The corporate sector pays numerous taxes, levies, and fees to the government. e-Taxation will make it easier to pay these taxes and levies. Online taxation and payment of taxes can help to reduce the cost and time required for physical tax submission. ICT can also help cross-check payment frauds and flaws, increasing accuracy and revenue for the government.
- e-Licensing: Companies must obtain multiple licences from the government, just as they must obtain various registrations. ICT enabled licencing and registration can save time and money.
- e-Tendering: e-tendering features online tendering and procurement capabilities. It would provide online notifications of new business possibilities with the government, as well as online tender filing and work allotment. It will save time and money in the formal tendering method.

SMART Governance

- SMART stands for simple, moral, accountable, responsive, responsible, and transparent governance.
- S: The use of ICT simplifies governance through electronic paperwork, online submission, and online service delivery, among other things.
- M: It provides morality to administration by eliminating immoralities such as bribery, red tapism, and so on.
- A: It holds the government's response by making all of the government's data and information available online for all citizens, NGOs, and the media to review.
- R: Government agencies become more responsive as a result of less paperwork, increased communication speeds, and decreased communication time.
- T: With higher morale, online information availability, and less red tape, the governing process becomes open, leaving little room for the government to withhold any information from citizens.

Advantages of e-Governance

- Speed: Technology has accelerated communication. The Internet, telephone, and smartphones have shortened the time required for normal communication. The majority of government spending is allocated to the expense of stationary. Paper based communication necessitates a large amount of stationary, computers, and so on, necessitating ongoing high investment. The web and smartphones make communication more affordable, saving the government money.
- Transparency: The usage of ICT makes the governance process more visible. All government information would be made public on the Internet. Citizens can obtain information whenever they want. However, this is only conceivable if all information about the government is posted on the internet and accessible to the public. The existing governance process provides numerous opportunities to conceal facts from all citizens. ICT assists in making information available online, eliminating any possibilities of information concealment.
- Accountability: When the governing process is made public, the government becomes accountable. Accountability refers to the government's responsibility to the people. It is the accountability for the actions of the government. A responsible government is accountable.

Objectives of e-Governance

- To create an informed society: A well informed society is an empowered community. Only knowledgeable citizens can hold a government accountable. One of the fundamental goals of e-governance is to provide everyone with access to all facts about the government and its public importance.
- To promote engagement between the government and citizens: In the real world, the government and citizens rarely interact.

The feedback given from and to citizens is insignificant. e-Governance strives to provide a feedback system, collect feedback from citizens, and make the system aware of citizens' problems.

- Promote citizen involvement: True democracy necessitates the participation of every citizen. Increased population has resulted in a representational democracy, which is not a true democracy. e-Governance attempts to restore democracy to its real meaning by boosting citizens' participation in the governing process, including feedback, access to data, and overall citizen participation in decision-making.
- To increase openness in the governing process: e-Governance aims to increase transparency in the governing process by making all government data available to the public. Its purpose is to inform the public about the government's actions and policies.
- Hold the government accountable: The government is accountable for every action or decision it makes. e-governance strives to make the government more accountable than it is presented through increasing transparency and informing citizens.
- To decrease the cost of governance: e-governance aims to reduce the cost of governance by reducing expenditure on physical delivery of information and services: It seeks to achieve this by reducing expenditure on stationary, which accounts for the majority of the government's budget. It also eliminates the need for physical communication, lowering the time and expense of communication.
- To decrease the government's reaction time: Normally, the government takes a while to respond to people's questions and problems due to red tape and other factors. Because problems are government's problems, as government is for the people, e-governance aspires to minimise the government's reaction time to the people's queries and complaints.

Stages of e-Governance

- The United Nations e-readiness survey takes into account the following phases of e-governance in terms of governments' online presence worldwide.
- Developing presence: The stage l, or emerging presence, takes into account the online availability of minimal and basic information. A basic e-government online presence consists of an official site. Links to central government ministries and departments, as well as regional/local government, may or may not be available. At this point, the website may also contain archive information such as messages from heads of state or the constitution. Nevertheless, at this point, much of the information is static, with no options for citizens.
- Enhanced presence: The UN considers stage II to be the government's enhanced online presence. At this point, the government offers additional public information resources including government policies, laws, guidelines, reports, and newsletters. This could potentially be downloaded at this point. This stage may allow the user to search for information within online documents. A sitemap function may also be included on the webpage website to navigate easier. Although there are more materials available online at this point, the website's navigation is still basic and unidirectional. At this point, there is no interactiveness because the information is solely going towards citizens rather than from people to the government.
- Interactive presence: In stage-III of Interactive presence, the governments are encouraged to implement interactivity on their websites. It entails the availability of government online services to improve consumer convenience. This will feature downloadable forms and applications for bill payment, taxes, and licence renewal.
- At this point, government websites would feature audio and video capabilities to promote citizen interaction. At this point, government authorities can be contacted via email, fax, phone, and postal mail. The website would be regularly updated to keep the public informed and up to date.
- Transactional presence: Transactional presence allows citizens and the government to interact in two ways. It allows citizens to

submit forms and applications online 24/7, allowing them to pay taxes, and apply for ID cards, birth certificates, passports, license renewals, and other similar C2G activities.

- At this point, citizens will be able to use their credit, or debit card to pay for applicable public services such as traffic violations, taxes, and postal service fees. Suppliers of products and services can compete for public contacts online via secure links.
- Networked presence: According to the UN Survey, stage V is 'networked presence,' which constitutes the most sophisticated stage in online e-government activities. It is distinguished by the combination of G2G, G2C, and C2G interactions. At this level, the government welcomes citizen participation and online participatory decision-making, and it is willing and equipped to engage society in a two-way open discussion.
- The government will actively solicit citizens' comments on public policy, law-making, and democracy participatory decision-making through interactive content such as the web comment form and novel online consultation techniques. The integration of public sector institutions with complete collaboration and knowledge of the concept of collective decision-making, participatory democracy, and citizen empowerment as a constitutional choice is launched at this level of e-governance.

Strategies for e-Governance in India

- To create technical infrastructure/frameworks throughout India: India needs a full-fledged ICT framework for e-governance implementation. Complete e-governance deployment in India will require the development of technical hardware and software infrastructure. It will also have improved and faster networking choices. Faster broadband services and faster wireless networks, such as 3G and 4G, will be among the newer connectivity possibilities.
- Infrastructure must be constructed by the government, private sector, and individuals. The infrastructure will also promote Cyber Cafes, Knowledge, and Interactive Kiosks. However, disabled people must be considered when constructing technical infrastructure. The technology that is implemented must include disabled people.
- Building institutional capacity: In addition to technical facilities, the government must establish institutional capability. This will include coaching for government employees as well as the hiring of experts. The government must also build an Expert database to properly utilise intellectual resources. Aside from that, the government must equip agencies with cutting edge technology and establish a special investigative agency.
- To construct legal infrastructure: To better execute e-governance, the government will need to draught laws that completely embrace both established and emerging technologies. Many preconceived conceptions have been altered by evolving technology; similarly, technology is rapidly growing and changing. The government must enact regulations that embrace current technology while also leaving room for future technology to evolve. These IT rules must be adaptable to keep up with quickly evolving technology. India now has only the IT Act of 2000, which is primarily e-commerce legislation. India has also amended numerous laws to accommodate electronic technology; nonetheless, this is insufficient to fully include e-governance.
- To create judicial infrastructure: Current judges have a relatively low level of technology awareness. The judicial system as a whole needs to be educated on modern technology, its advantages and disadvantages, and its varied applications. Alternatively, the judiciary may appoint new judges and establish Special Courts to address ICT related issues. The government may even establish special tribunals to handle ICT related issues.
- Make all data available online: The government needs to make all information available on the Internet via websites. This can be aided by centralised information storage, content localisation, and content management. Because government information is public, citizens have the right to know everything there is to know about the government, because

the government is run by the people, for the people, and the people.

- Why popularise e-governance? India's literacy rate is worrying. The entire globe is shifting to e-government, yet India still requires a literacy department. For e-governance to thrive, people must be educated and become e-literate. In India, there are extremely few e-literate people.
- The government must promote e-governance and raise public knowledge of its benefits. The government can only incentivise people to go online if it makes them feel at ease with e-governance. This can be accomplished by educating the public about the benefits of e-governance over physical governance. This can also be accomplished by increasing the awareness of leaders who can strongly urge citizens to go online.
- Centre state partnership: The Indian system is semi federal. As a result, inter- and intrastate collaboration is required for the democratic process to run smoothly. This collaboration is also required for successful e-governance deployment. This collaboration will extend to centre state, interstate, and interdepartmental ties. For this purpose, the government can establish a Federal Hub, similar to the existing Government of India website, to allow access to information from all central government organs as well as state governments. The states and the centre can work together to build a National Citizen Database.
- Establishing standards: Finally, it is critical to establish various standards to bring e-governance up to the quality and reliability level of the private business world. The government of India is presently focusing on norms, and administration and has several draughts ready. There is currently no universal standard for government websites in India. Many Maharashtra government websites have standards that differ between even two of their internet pages. There is no fixed standard for the quality of information, documents, formats, and so on. The government should establish universal national standards that all governments and agencies must follow.

M-Governance

- M-governance does not replace e-government; rather, it supplements it. M-governance refers to the use of mobile or wireless technology to improve governance, services, and information 'anytime, anywhere.' Strong back ICT infrastructure and workflows are also required for mobile applications. It has the potential to use smartphones as input devices in specific regions where the last mile connection is a difficulty for simple data inputs vital to government decision-making.
- M-governance is not a new thought. The private sector has made extensive use of mobile phones to provide value added services for the following, which are largely SMS based banking, media, airlines, telecommunications, recreation, news, sports, horoscope, movie tickets, and so on.
- M-governance has raised the efficiency of public servants, improved the flow of services and information, expanded avenues for public involvement, and reduced expenses, resulting in greater public participation.

Mobile Seva

- It intends to provide citizens with government services via smartphones and tablets. It was created as the foundational architecture for making public services available via mobile devices.
- Mobile seva allows the mobile platform to be integrated with the common e-government infrastructure, which includes state wide area networks (SWANs), state data centres (SDCs), and state and national service delivery gateways (SDGs or NSDG).
- It allows a government body to seamlessly integrate web and mobile based services and vastly improves access to electronic services by utilising the extremely high adoption of mobile phones, particularly in rural areas.

Challenges in e-Governance

- There are numerous possible hurdles to e-governance deployment. Some barriers to implementation include security, unequal

citizen access to computer technology, the high initial cost of putting up e-government systems, and aversion to change. Trust, reluctance to change, the digital gap, cost, privacy concerns, and security concerns were mentioned as challenges.

- Trust: trust can be defined in two ways as a current circumstance judgement or as an intrinsic personality and inclination. The use of e-government to carry out the activities of public administration necessitates the existence of two levels of trust. The first requirement is that the user is confident, at ease, and trusting in the instrument or technology with that they will interact. The second aspect of trust is trust in the government.
- There must be a balance struck between assuring that a system avoids fraudulent activity and the burden placed on honest people by thorough checks.
- When a computer storing their personal information was stolen recently, confidential information about military veterans was jeopardised. This type of occurrence has the potential to undermine trust and user faith in government systems. Trust, coupled with financial security, is a significant concern limiting e-government service uptake.
- Resistance to change: According to the innovation diffusion theory, over time, innovation will disseminate through a population, with the pace of acceptance varying between those who adopt early, known as early adopters, and others who embrace the innovation much later, known as lagers.
- Citizens, employees, and corporations might all have preferences as to how transactions should be handled. Government institutions and public policy managers, on the other hand, cannot ignore the changes brought about by the use of communication and information technology. One approach toward decreasing some of the existing resistance is education about the value of the new systems. It can also be especially beneficial for a leader or manager to engage in the new system early in the adoption process.
- Digital divide: The digital divide is the gap that occurs between individuals, organisations, and enterprises that have access to technology and those that do not.
- The presence of the digital gap can be explained by social, economic, infrastructure, and ethnolinguistic characteristics. Why a person living below the poverty line cannot afford a computer to make use of e-government and other Internet services. As the digital divide closes, more people will be able to use e-government in the public sector. The digital gap is caused by factors other than economic poor. It also can be caused by a lack of public knowledge. Even some financially secure people are unaware of the reach of e-governance.
- Awareness can only assist consumers to find a delivery services channel once. It cannot guarantee continued utilisation unless the system is also intended to produce an acceptable result. Procedures must be streamlined to provide meaningful benefits, and clear directions must be provided to facilitate their use by actual end users and eliminate reliance on middlemen/intermediaries.
- Cost: One of the most significant impediments to e-governance deployment is cost, mainly in developing nations like India where the majority of the population lives below the poverty line. Elected officials and legislators appear uninterested in implementing e-governance.
- Privacy and security: For e-government stakeholders, there will be three basic levels of access: no access to a Web service, limited access to a Web service, and full access to a Web service; however, when personal sensitive data exists, the establishment of the security access policy is a far more complex process requiring legal consideration. Effective steps must be taken to protect personal data while implementing e-government projects. The implementation of projects including confidential material such as income and medical history might be hindered by the lack of security screening standards and practises.

Disadvantages of e-Governance

- The biggest disadvantage of electronic governance is the transition of government services to an electronic based structure. This technique eliminates the person to person interaction that many individuals appreciate.
- Furthermore, with several technology based services, it can be easy to create the excuse (e.g., the server has taken a hit) that difficulties with the service offered are due to the technology.
- There are several limitations to implementing e-governance. Users' literacy and computer skills; users who may not read or understand or write will require assistance. Senior citizens are a good example. In general, senior adults lack computer literacy and would need to seek assistance from a customer service representative. The rural area populations also provide opportunities for middlemen to falsify information.
- Studies have revealed that there is a risk of a decrease in the accessibility of government online attributable to factors such as access to Internet technology, service usability, and computer access.
- Even though the public has a high level of trust in the security provided by government websites, the public is nevertheless concerned about security, spam from disclosing email addresses, and government preservation of transactions or user behaviour. Concerns concerning the privacy of data obtained as part of the UID programme have grown. The security of cyberspace and data exploitation continue to prevent citizens from fully embracing the Aadhar card.

Digital India Programme

- The Government of India's flagship programme, Digital India, aims to turn India into a digitally enabled nation and knowledge economy.
- Digital India's vision: the program's objective is to convert India into a technology driven country and knowledge economy.
- Digital India vision areas: the digital India initiative is focused on three core vision areas.
    - Digital infrastructure as a functionality for every citizen
    - Governance and on demand services
    - Citizen digital empowerment

Infrastructure as a Service for Every Citizen

High-speed internet is a key utility. Unique, everlasting, online, genuine digital identity from birth to death. Mobile phone and bank account for involvement in digital and financial space. Convenient access to a common service centre. Accessible private space on a public cloud.

Governance and Services on Demand

Services are accessible in real from online and mobile platforms. All citizen welfare benefits are to be available on the cloud. Services digitally transformed to improve the ease of doing business. Making banking transactions electronic and cashless. Utilizing GIS for decision support systems and development.

Digital Empowerment of Citizens

Universal digital literacy, universally accessible digital resources, all documents/certificates to be available on the cloud, provision of digital resources/ services in Indian languages, collaborative digital platforms for participatory governance, portability of all entitlements via cloud.

Pillars of Digital India

- Digital India is a programme that covers numerous Government Ministries and Departments. It binds a great number of ideas and thoughts into a unified, comprehensive vision, from which each can be achieved as part of a bigger purpose. Each component stands alone but also contributes to the overall picture. Digital India will be implemented by the whole government, with the Ministry of Electronics and Information Technology (MeitY) serving as the overarching coordinator.
- Digital India intends to give the nine cornerstones of growth a much needed boost. They are broadband highways, universal access to mobile connectivity, public internet access

programme, e-governance: revolutionising government through technology, e-Kranti—electronic delivery of services, information for all, electronics manufacturing, IT for jobs, and early harvest programmes.

- Each of these sectors is a complex programme in and of itself, spanning several Ministries and Departments.
- Broadband highways: This component has three sub-components: broadband for all-rural, broadband for all-urban, and national information infrastructure (NIl).
- Wireless broadband for the entire rural area: By December 2016, the national optical fibre network (NOFN) would cover 250000 local panchayats. The department of telecommunications (DoT) is the project's nodal agency.
- All urban broadband: virtual network operators would be used for the delivery of services and telecommunications networks in new urban development and buildings.
- National information infrastructure (NII): The NII would integrate the country's network and cloud infrastructure to give high-speed connectivity and a cloud platform to various government ministries down to the panchayat level. Networks such as the State-Wide Area Network (SWAN), the National Knowledge Network (NKN), the National Optical Fibre Network (NOFN), the Government User Network (GUN), and the MeghRaj cloud are among the infrastructural components.
- Universal access to mobile connection: This major campaign deals with network penetration and closing connectivity gaps throughout the country.
- There are approximately 55619 communities in the country that lack cellphone connectivity. The provision of mobile service to unconnected communities has begun as part of the North East's comprehensive development strategy. Mobile connectivity to the remaining unconnected villages would be phased in.
- Public internet access programme: The public internet access programme has two sub components: common services centres (CSCs) and post offices as multi service facilities.
- Common services centres (CSCs): CSCs would be strengthened and increased in number to 250000, with one CSC in each gram panchayat. CSCs would be transformed into viable, multifunctional end nodes for the provision of business and government services. The scheme's implementation would be overseen by the deitY.
- Post offices as multi service centres: It is recommended that 150000 post offices be turned into multi service centres. The Department of Posts would be the agency responsible for carrying out this scheme.
- e-Governance: using technology to reform government: the political process re-engineering utilising IT to simplify and improve government processes is vital for transformation to improve the delivery of public services across multiple government domains and must thus be undertaken by all Ministries/Departments.
- e-Kranti (National e-Governance Plan 2.0): Electronic service delivery: On May 18, 2006, the government adopted the National e-Governance Plan 1.0 (NeGP1.0), which included 31 mission mode projects (MMPs) and eight components.
- e-Kranti (NeGP 2.0) is a critical component of the digital India plan. Given the essential need for e-governance, mobile overnance, and good governance in the country, the Union Cabinet adopted the strategy and major components of e-Kranti on March 25, 2015, with the aim of "transforming e-governance for transforming governance." All new and continuing e-governance projects, as well as old programmes that are being updated, should now adhere to the e-Kranti main principles. e-Kranti currently has 44 projects at varying phases of development.
- General information: The open data platform enables ministries/departments to proactively release datasets in an open format for use, repurposing, and redistribution. Citizens would benefit from open and simple access to information if information and papers were hosted online.

- The government must participate in proactive citizen engagement through social media and internet platforms. MyGov.in, a forum for citizen participation in governance, as a channel for exchanging ideas/suggestions with the government.
- Electronics manufacturing: This pillar focuses on boosting electronics production in the country, with the goal of achieving NET ZERO imports by 2020. Targeting NET ZERO imports is a powerful statement of purpose. This lofty aim necessitates coordinated action on multiple fronts, including taxation, incentives, economies of scale, elimination of cost disadvantages, priority areas ticket items, incubators, clusters, skill development, PhD enhancement, government procurement, safety standards-compulsory registration, assistance for labs and MSMEs, national award, marketing, brand building of national centres-flexible electronics, security forces, and R&D in electronics.
- Demand for electronic goods is rising at a 22 per cent compound annual growth rate (CAGR) and is estimated to reach $400 billion by 2020. The Indian government is taking many initiatives to boost production and investment in this industry, putting India at the top of the list of potential investment destinations.

IT for Jobs

This section focuses on teaching youth the skills necessary for job prospects in the IT/ITES sector. This pillar includes eight components, each with its own set of activities.

IT Training for People in Smaller Towns and Villages

The goal of this component is to teach one million students from small towns and villages about IT careers during the next 5 years. DeitY is the scheme's nodal department.

IT in the North-Eastern States

This component focuses on establishing BPOs in every north-eastern state to assist ICT enabled development in these states. DeitY is the scheme's nodal department.

Training Service Delivery Agents

The focus is on educating 300000 service delivery agents as part of professional development to manage viable enterprises delivering IT services. DeitY is the scheme's nodal department.

Telecom and Telecom Related Training for Rural Workers

This component focuses on training 500000 rural workforce telecom service providers (TSPs) to meet their demands. This scheme's nodal department is the department of telecommunications (DoT).

North-East BPO Promotion Scheme (NEBPS)

- India has steadily emerged as a preferred BPO destination around the world. Several factors, including operational cost-effectiveness, the availability of skilled labour, and the ever-increasing need for employment possibilities, have all contributed to the country's BPO industry's expansion. The BPO industry, on the other hand, has been predominantly centred in and near big (Tier I) cities.
- The recurring labour cost to the organisation is believed to be higher in large (Tier I) cities, especially given the considerably higher cost of housing units and longer commute distance for employees. As a result, a BPO company would be wise to relocate to smaller (Tier II/III) cities, particularly those in the North Eastern Region, as this would result in much lower manpower related expenses, making its operations far more profitable.

Early Harvest Programmes

- The early harvest programme comprises initiatives that must be completed in a short period. The following projects are part of the early harvest programme: messages IT platform, government greetings will be replaced by e-greetings, biometric attendance, Wi-Fi in all universities, secure email within government, streamlined government email

design, public Wi-Fi hotspots, school books to be e-books, SMS-based weather and disaster alerts, and national portal for lost and found children.

- Digital India services: Some of the services that will be supplied through this programme include a digital locker, e-education, e-health, e-sign, and a national scholarship portal.
- The digital locker service would allow citizens to digitally store critical papers like PAN cards, passports, mark sheets, and degree certificates. The digital locker will help facilitate access to government papers. It takes advantage of Aadhaar's authentication services. Its goal is to eliminate the use of hard copies while allowing government entities to share verified electronic records.
- The national scholarship portal is a one-stop shop for the whole scholarship process, from student application to verification, sanction, and disbursement to the end beneficiary, for all scholarships offered by the Government of India.

Government of India Initiatives

National-Governance Action Plan (NeGP 1.0)

- India's national e-governance action plan tries to solve many of the key concerns of e-governance for the average citizen.
- The NeGP (2003–07) structure consists of a collection of core policies for unification and assistance, a set of integrated projects or cross cutting efforts, and a set of mission mode projects at the national and state levels.
- NeGP's vision is to "make all government services accessible to the common man in his locality, through common service delivery outlets, and to assure the efficiency, transparency, and reliability of such services at affordable costs to realise the basic needs of the common man."
- On 18 May, 2006, the government adopted the e-governance plan, which included 27 mission mode projects (MMPS), eight components, three essential components, and five others.
- Mission Mode Projects (MMPs): The figure below displays the NeGP-8's 27 MMPs, which include eight integrated projects, eight central projects, and 11 state projects.

The Eight Integrated Projects

- e-BIZ aims to address numerous difficulties linked to business approvals and permissions, such as minimising the points of interaction between corporate entities and government authorities, standardising necessary information, enabling a single window service, and lowering the compliance burden.
- The national service delivery gateway functions as a messaging middleware, intelligently directing services from a service consumer to a service provider.
- Common service facilities intends to establish 84.419 centres in 17 states to provide common services.
- e-Procurement provides a one stop shop for all government procurement services.
- e-Office offers an interface for streamlining, aligning, optimising, and automating all internal government activities.
- e-Courts provides online access to decisions and cause lists, as well as case filing and notice by email.
- The India portal is a one-stop shop for government services.
- Electronic data interchange (EDI) for trade incorporates electronic filing and authorisation of import and export records, e-payment of duties and charges by ports, airports, customs, and other government agencies, and electronic exchange of documents among partners agencies and customs, ports, and other government entities.

The Eight Central Projects

- Income Tax comprises 19 defined online services such as e-filing of tax returns via intermediaries, online filing of forms, online payment of taxes through selected firms, refunds issued through the electronic clearance system, and so on.
- MCA21 provides access to all ministry of company affairs (MCA) services, including document filing, company registration, and

public access to business information, via a secure online.
- The four government service general insurance companies provide insurance related services.
- Central excise allows for the submission of service tax and excise returns, as well as e-payment of custom taxes. Courier consignments are cleared automatically, for example.
- National ID/UID provides a centralised database and creates unique IDs for residents from across the country, largely to ensure that social sector benefits are delivered effectively.
- Pensions monitors and manages pension settlements.
- Banking connects several banks' essential banking solutions.
- Passports, visas, and immigration allow people to apply for new passports and renew old ones. All international arriving flights' application status is tracked, and immigration requirements are handled.

The 11 State Projects

- Agriculture provides services such as market prices, soil knowledge, crop disease and management, appropriate horticultural and sericulture practises, and so on.
- Employment exchange allows employers and potential employees to match their needs. Employers can post vacancies online and receive guidance from the unemployed.
- Commercial taxes increase the effectiveness of VAT administration by allowing e-filing of returns and refund clearance, online tax payment, etc.
- Land records identifies and automates 14 services, such as the integration of textual and geographic land records data, the integration of registration and mutation procedures, the automatic updating of land records that provides irrefutable title to landowners, and so on.
- Road transport services include car registration and driving licenses.
- Gram panchyat issues trade licenses, certificates, house related services, fund status reports, individual recipients of various schemes, and so forth.
- Municipalities offer G2C services such as birth and death certificates, utility bill payments, licence issuance, and so on.
- In police stations, officers use the common integrated police application (CIPA) and hardware.
- Property registration replaces the manual method of document verification and security, document capture and preservation, document searches, and back office record maintenance.
- Treasuries are used to pay government employees' wages, expenditures, and so forth.
- e-District automates backend operations to enable G2C service delivery via common services centres. Each state may add up to five state specific projects in addition to the 11 programmes defined.

Information Technology Act, 2008

- This act may be referred to as the information technology act of 2000. [As amended by the information technology (amendment) act, 2008], H applies to the entire country and, unless as otherwise specified in this act, it also applies to any offence or contravention committed outside India by any person.
- It is an act to provide official status for transactions conducted using electronic data interchange and other forms of electronic communication, colloquially known as 'electronic commerce,' which incorporates the use of alternative solutions to paper based methods of communication and information storage to facilitate e-filing of documents with government agencies.
- Highlights of points and some outcomes are provided below about IT Act 2000.
    - 'Certifying authority' refers to a person who has been granted a licence to produce an electronic signature certificate.
    - 'Digital signature' refers to a subscriber's validation of any electronic record using an electronic technique.
    - 'Cybersecurity' is preventing unauthorised access, use, disclosure, disruption, alteration, or destruction of information, equipment, devices, computers, computer

resources, communication devices, and information held therein.
    - 'Electronic signature' refers to a subscriber's verification of any electronic record using the electronic technique indicated in the second schedule, which includes a digital signature.
    - In an asymmetric cryptosystem, a 'key pair' is a private key and its mathematical associated public key, which are so connected that the public key may validate a digital signature issued by the private key.

Cyber Appellate Tribunal (Amended vide ITA-2008)

Members of the Cyber Appellate Tribunal (Substituted Vide ITAA 2008)

- The cyber appellate tribunal shall be composed of a chairperson and such additional members as the central government may appoint by notification in the official gazette (Inserted vide ITAA-2008).
- The central government will choose the chairperson and members of the cyber appellate tribunal in collaboration with the chief justice of India. (Inserted in accordance with ITAA-2008).
- Qualifications for Chairperson and Members of the Cyber Appellate Tribunal (Substituted vide ITAA-2006).
- No one shall be eligible to serve as a Chairperson of the Cyber Appellate Tribunal unless he is, has been, or is competent to be a High Court Judge (substituted vide ITAA-2008).
- Except for the Judicial Member to be appointed under subsection (3), the Members of the Cyber Appellate Tribunal shall be appointed by the Central Government from among persons with special knowledge of and professional experience in information technology, telecommunications, industry, management, or consumer affairs.

Chairperson and Members' Terms of office, Terms of Service, and So On (As per ITAA-2008, substituted)

- The Chairperson or Member of the Cyber Appellate Tribunal shall serve for a term of five years from the date of his appointment or until he reaches the age of 65 years, whichever comes first. (Inserted in accordance with ITAA-2008).
- Before appointing someone as the Chairperson or a Member of the Cyber Appellate Tribunal, the Central Government must be satisfied that the person has no financial or other interests that would jeopardise his or her powers as such Chairperson or Member. (Inserted in accordance with ITAA 2008).
- An officer of the Central Government or a State Government who is appointed as the Chairperson or a Member of the Cyber Appellate Tribunal, as the situation may be, must retire from service before taking up the position of Chairperson or Member. (Inserted in accordance with ITAA 2008).

e-Kranti (Electronic Delivery of Services)

- e-Kranti is a critical component of the Digital India initiative. e-Vision Kranti's is transforming e-governance for transforming governance.
- The objective of e-Kranti is to enable a government wide transformation by delivering all government services digitally to citizens via integrated and interoperable systems through numerous channels while assuring efficiency, accountability, and reliability of such services at affordable rates.
- The following are the goals of "e-Kranti."
    - To reimagine NeGP through revolutionary and outcome driven e-government activities
    - Expanding the portfolio of citizen centred services
    - Ensure optimal utilisation of essential ICT
    - To encourage the quick duplication and integration of e-government applications
    - Making use of developing technology
    - Using more agile implementation paradigms
- The following are e-Kranti's principles.
    - Transformation rather than translation
    - Integrated services rather than individual services
    - Government process re-engineering (GPR) will be required in all MMPs

- On demand ICT infrastructure
- Cloud by design
- Priority for mobile
- Approvals are expedited
- Imposing standards and protocols
- Language translation
- National geographic information system (geospatial information system)
- Security and electronic data storage

Digital India Initiatives

Digital Locker System

- The digital locker is an important component of the digital India initiative. The goal of digital locker is to reduce the use of physical papers while allowing agencies to share e-documents. The e-documents would be distributed through registered repositories, assuring the online legitimacy of the records.
- Residents can also use the e-sign service to upload digital files and digitally sign them. These digitally signed documents can be disclosed to government agencies and other groups.
- The following benefits come with the digital locker system.
  - Print from anywhere: All digital records will be in a printable format, allowing users to print from anywhere at any time. This will enable customers to access the papers in the event of a calamity or document loss.
  - Convenience: There will be no longer long lines or procedures to obtain the documents.
  - Cost-free: The digital locker service is supplied to Indian residents at no cost, allowing it to be used by people from all socioeconomic backgrounds.
  - Shareable: Residents may simply exchange documents with other agencies/departments without the use of photocopies, scan copies, document uploads, and so on.

My Gov Mobile App

- My Gov Mobile App is a unique platform that uses technology to create a relationship between citizens and the government for India's growth and development. The government hopes to increase citizen engagement in good governance through this platform by soliciting ideas, proposals, and grassroots contributions.
- Citizens can participate in this unique nation building initiative, and for the first time in India's history, citizens from across the country will come together to share their expert ideas, thoughts, and recommendations with the government in aspects related to different policies, and programmes, schemes, and so on. MyGov will enable citizens to collaborate with the government.

e-Sign

- e-Sign allows an aadhaar holder to digitally sign a document utilising an online service. e-Sign is intended for use in digital signature applications that require signer authentication via the Aadhaar e-KYC service. This is a bundled service that enables the issuance of a digital signature certificate and the signing of required data by validating the aadhaar holder. Aadhaar is required to use the e-sign service.
- A simple and safe way to digitally sign material anywhere, at any time: e-sign is an Internet service that allows application service providers to verify signers and perform digital signatures of documents using the aadhaar e-KYC service. It is not necessary to use hardware tokens.
- Allows for legally acceptable signatures: the design process includes consumer consent, digital signature certificate generation, digital signature generation and affixing, and digital signature certificate acceptance following the provisions of the information technology (IT) Act of 2000. It ensures compliance through API specifications and API licencing models. A comprehensive digital audit report is also kept to check the validity of transactions.
- Configurable and simple to implement: e-sign provides configurable authentication settings in accordance with the aadhaar e-KYC service and also saves the aadhaar number that is

used to authenticate the signer's identity. e-KYC authentication methods include biometric (fingerprint or iris scan) and one-time password (OTP) (through the registered mobile in the Aadhaar database). Millions of aadhaar holders now have simple access to legally recognised digital signature services thanks to e-sign.

- Privacy: e-sign protects the signer's privacy by requiring only the fingerprint (hash) of the document to be submitted for the signature functionality rather than the entire document.
- e-Sign service is a secure online service governed by e-authentication criteria. While the signer is authenticated using aadhaar e-KYC services, the signing on the document is performed on an e-sign provider's backend server. A trustworthy third-party service provider, currently certifying authorities (CA) licenced under the IT Act, provides e-sign services. To improve security and avoid abuse, private keys for aadhaar holders are generated and destroyed on hardware security modules (HSM).

e-Hospital

- As part of the government of India's digital India effort, a hospital management system with a citizen facing interface has been established, providing a uniform platform for hospitals and citizens to access healthcare services. This portal seeks to: create a uniform forum for citizens to access healthcare services online; and establish patients' identities using their aadhaar numbers.
- The advantages are saving citizens' time and effort in obtaining healthcare services, providing health services at citizens' doorsteps, providing unique health identification (UHID) number for on boarded hospitals utilising aadhaar, and assisting hospitals in improving patient care.

Bharat Net

- The Bharat Net is the world's largest optical fibre based rural broadband connectivity project. It would provide a least 100 Mbps bandwidth to each of the 2.5 lakh gram panchayats. A high capacity network management system and a network optical centre will support this network.
- Bharat net will act as digital India's infrastructure backbone, bringing Internet and telecommunications services to every corner of the country. It would make it easier to provide e-governance, e-health, e-education, e-banking, and other services to rural India.

PRAGATI (Pro-Active Governance And Timely Implementation)

- PRAGATI is a one of a kind integrated and interactive platform. The platform aims to address common people's problems while also monitoring and reviewing key initiatives and programs of the government of India as well as those sponsored by state governments.
- The PRAGATI platform is unusual in that it combines three cutting edge technologies.
  - Digital data administration
  - Video conferencing
  - Geographical technology
- It also provides a unique opportunity for cooperative federalism by bringing together the secretaries of the government of India and the chief secretaries of the states on one stage.
- With this, the prime minister can debate matters with relevant central and state officials, armed with complete information and the most recent images of the ground-level situation. In India, such an endeavour has never been made. It is also unique e-governance and good governance project.
- The following are the key characteristics of the PRAGATI application.
  - It is a three tier system (PMO, union government secretaries, and state chief secretaries).
  - The prime minister will host a monthly programme in which he will interact with Government of India Secretaries and State Chief Secretaries via video conferencing enabled by data and geo informatics visuals.

- The first such programme was introduced on 25 March 2015 (Wednesday) at 3:30 p.m.
- From now on, it will be held once a month on the fourth Wednesday at 3:30 p.m., dubbed PRAGATI day.
- Issues to be brought before the PM are selected from a database of public grievances, ongoing programmes, and impending projects.
- The system will be built on, strengthened, and re-engineered datasets from the CPGRAMS for grievances, the project monitoring group (PMG), and the ministry of statistics and program implementation. PRAGATI provides an interface and platform for all three aspects.
- It will also consider various correspondences to the PM's office from ordinary citizens or high dignitaries of states and/or developers of public projects.
- The issues reported are posted 7 days before the PRAGATI day (i.e., on the third Wednesday of every month).
- After entering the application, Union Government Secretaries and Chief Secretaries can view these issues.
- User ID and Password have been created and made available for each of the Union Government Secretaries and Chief Secretaries.
- Union Government Secretaries and Chief Secretaries will be allowed to see the issues of their Department/State.
- Union Government Secretaries and Chief Secretaries must provide comments and updates on the flagged issues within 3 days (i.e., by next Monday).
- The PMO team has one day (Tuesday)—available to examine the data entered by the Union Government Secretaries and Chief Secretaries.
- The design is such that when the PM analyses the issue, he should have the issue as well as the latest updates and graphics on his screen.
- The system was created in-house by the PMO team with assistance from the National Informatics Centre (NIC). Its goal, as the name implies, is to establish a culture of proactive governance and timely implementation. It is also a strong mechanism for delivering e-transparency and e-accountability to important stakeholders through real-time presence and exchange.

Meghraj (Gi Cloud)

- The GI cloud will be made up of many national and state clouds. The authorities in charge of operating and managing the national and state clouds may contract with managed service providers (MSPs) to manage the cloud computing environments. These cloud computing environments will make use of existing network infrastructure like SWANs, NKN, NOFN integration hubs, and internet. It is carried out by NIC.
- The following are the goals of GI cloud: optimal infrastructure utilisation, accelerating the development and deployment of eGov applications, easy replication of successful applications across states to avoid redundancy of effort and cost in the development of similar applications, and availability of certified applications that adhere to common standards in one location. It assists federal departments with cloud computing.

National Supercomputing Mission

- The Mission envisions strengthening our nation's academic and research institutions across the country by constructing a massive supercomputing grid with more than 70 high-performance computing (HPC) centres.
- These supercomputers will be linked via the National Knowledge Network (NKN) to the national supercomputing grid. The NKN is another government programme that links academic institutions and R&D labs over a high-speed network. Academic and research institutions, as well as major user departments/ministries, would take part by utilising these resources and producing applications of national significance. The

mission also includes the creation of highly skilled high-performance computing (HPC professionals to handle the challenges of developing these applications.

- The mission implementation will bring supercomputing within the grasp of the country's enormous scientific and technological community, enabling the country to solve multidisciplinary grand challenge challenges.
- The mission would be undertaken and managed jointly by the Department of Science and Technology—(DST) and the Ministry of Electronics and Information Technology (MeitY) at an anticipated cost of Rs 4500 crore over seven years.
- To make India a world leader in supercomputing and to strengthen India's capability in tackling grand challenge problems of national and global importance.
- To provide our scientists and researchers with cutting edge supercomputing facilities so that they can conduct cutting edge research in their respective domains.
- To achieve global competitiveness and self-reliance in the strategic field of supercomputing technology by minimising repetitions and duplication of efforts and optimising investments in supercomputing.

Digital Payment Method

The digital India programme is the Government of India's flagship programme, to transform India into a digitally enabled society and knowledge economy. One of Digital India's declared roles is to be faceless, paperless, and cashless. Various ways of electronic payments are available to promote contactless payments and convert India into a less cash society. Some of them are banking cards, USSD, AEPS, UPI, mobile wallets, banks prepaid, point of sale (POS), internet banking, mobile banking, micro ATMs, bharat QR, etc.

USSD

Data from unstructured supplemental services. It is a new payment service *99#. It operates on a supplement data channel that is unstructured. It is suitable for mobile banking transactions utilising basic mobile phone features. There is no requirement for a mobile internet data connection for USSD based mobile banking. It is intended to give financial depth and inclusion to the underbanked populace in conventional banking services. The g9# service has been launched to bring financial services to every ordinary citizen around the country. It is a standard number used by all telecom service providers (TSPs) on their mobile phone to transact via an interactive menu presented on the mobile screen. *99# services include inter account to account transfer, balance query, and mini statement, among others. It is provided by 51 major banks and all GSM service providers and is available in 12 different languages. It operates using mobile money identifier (MMID) and mobile pin (MPIN).

Internet Banking

Online banking, e-banking, and online banking are an electronic payment system that allows customers of banks or other financial institutions to undertake a variety of financial transactions through the website of the financial institution.

National Electronic Fund Transfer (NEFT)

It is a countrywide payment system that enables one-to-one money transfers. Individual firms and corporate bodies can electronically transfer funds from any bank branch to any other bank branch in the country participating in the scheme under this scheme. Individuals who do not have a bank account can also deposit cash at NEFT enabled branches with guidelines to transfer funds using NEFT. Currently, NEFT runs on an hourly basis, with 12 settlements per day between 8:00 a.m. and 6:30 p.m. on weekdays and 8:00 a.m. to 12:00 p.m. on Saturdays.

Real-Time Gross Settlements (RTGS)

- It is a real time, continuous settlement of individual fund transfers on an order by order basis. Real time indicates that instructions are processed as soon as they are received rather than later. A gross settlement happens when each funds transfer instruction is settled individually.
- RTGS stands for real-time gross settlement. The minimum amount that can be transferred

via RTGS is 200000. There is no maximum amount for RTGS transactions.

Electronic Clearing System (ECS)

It is an alternative technique for completing payment transactions for utility bills, phone bills, insurance premiums, card payments, and debt payments.

Immediate Payment Service (IMPS)

It provides a 24 hours interbank electronic fund transfer service using mobile phones. IMPS is a powerful tool for immediately transferring money between banks in India via mobile, Internet, and ATM, which is not only secure but also cost-effective from both a financial and non financial standpoint.

Micro ATM

- A gadget used by business correspondents (BC) to provide basic banking services.
- The platform will allow BC to perform real-time transactions. The micro platform will function via low-cost devices (micro ATMs) linked to banks across the country. This could allow a person to deposit or withdraw money immediately. It will be powered by a mobile phone connection and made available to all BC. Customers would just need to have their identities verified before withdrawing or depositing funds into their bank accounts. This money will come from the BC's cash drawer.
- BC will operate as a bank for customers, and all they have to do is validate a customer's legitimacy using the customer's UID.
- Micro ATM transactions include deposit, withdrawal, fund transfer, and balance enquiry.

Point of Sale (POS)

- This is the location where sales are made. A point of sale (PoS) might be small, a market, or a metropolis on a micro level. A PoS, as defined by merchants, is the place where a customer performs a transaction, such as a checkout. It is sometimes referred to as a point of sale.
- The necessary conditions for PoS are a handheld device with a card or biometric reader, a merchant bank account, and GPRS/ Landline Internet access.
- Mobile PoS: requirements for mobile PoS are smartphone, bank app, integrated or external card, or biometric reader, the reader connects by a jack or bluetooth, internet connectivity (2G/3G/4G or WIFI), reader for QR codes or bar codes.

Payment Banks

- The RBI has sanctioned payment banks for 11 organisations, including major corporations such as Reliance Industries. Airtel and Vodafone, as well as Aditya Birla Nuvo and Tech Mahindra. These are the new, simplified banks that are expected to reach customers primarily through smartphones rather than regular bank locations.
- They cannot make loans, but they can accept deposits of up to Rs 100000 and pay interest on these balances in the same way as a savings bank account does.
- They can facilitate transactions and remittances by mobile phone.
- They can issue debit cards and ATM cards that can be used on any bank's ATM network.
- They can provide services such as automatic bill payment and buy in cashless, chequeless transactions over the phone. Because they are a part of the gateway that connects banks, they can move money directly to bank accounts at almost no cost.
- They can issue travellers a forex card, which can be used as a debit or ATM card throughout India.
- They can provide force services at a lesser cost than banks. They can also provide third party card acceptance mechanisms, such as Apple Pay.
- It is a significant step toward increasing financial inclusion in the country, as well as a step toward redefining banking in India. The RBI anticipates that payment banks will target India's migrant labourers, low income families, and small companies, offering low-cost savings accounts and remittance services. It expects that payment banks will allow poorer

residents who solely interact in cash to take this initial step toward formal banking.

- Traditional banks may find it uneconomical to open branches in every village, but mobile phone coverage offers a promising low-cost foundation for quickly bringing basic banking services to every rural resident. The innovation is also expected to hasten India's transition to a paperless economy.
- It will tackle India's domestic remittance sector, which is worth over $800 billion and increasing. It is tremendously beneficial to migrant workers. Payment banks can also help the government execute direct benefit transfers, in which subsidies for healthcare, education, and gasoline are transferred directly to the beneficiary's account.

Bharat QR

- With the launch of Bharat QR, the world's first interoperable payment acceptance solution, India has reached a new milestone in cashless transactions. This is the centre's second major programme, following BHIM, to boost digital transactions.
- A rapid response code (QR code) is a two-dimensional machine readable unique pattern that provides task specific information. A QR code, for example, is a newspaper advertisement or billboard that, when scanned with a smartphone, leads you to the advertiser's website.
- In retail payments, a QR code removes the hassle for a Point of Sale (PoS) or card swipe machine to complete electronic transactions. If the merchant and the consumer are both using the same payment app, such as PayTm or a bank app, the client can make the payment by scanning the merchant's QR code.
- The Bharat QR code is a payment acceptance solution that is interoperable. Even if the merchant and the client use different payment platforms, Bharat QR allows payment from one to the other. As a result, shops must show only one QR code rather than many ones for various payment platforms. It was created in collaboration with the National Payment Corporation of India (NPCI), Mastercard, and Visa.
- All NPCI affiliated banks are already operationally prepared to implement Bharat QR. This comprises. SBI, Axis Bank, Bank of Baroda, Bank of India, Citi Union Bank, Development Credit Bank, Karur Vysya Bank, HDFC Bank, ICICI, IDBI, RBL Bank, Union Bank of India Vlyaya Banka, and Yes Bank are among the financial institutions. Bharat QR can also be used by people who use Maser Using, Visa, American Express, or Rupay.

e-Governance Initiatives Are Taken by the Government

- The government of India's ministry of land resources has begun to digitise land records.
- AP and Gujarat e-procurement project: to minimise the effort and cost of doing business for both suppliers and the government.
- MCA 21: Issued by the Ministry of Corporate Affairs. It seeks to provide corporates and other stakeholders with quick and secure web access to all registry related services offered by the Union Ministry of Corporate Affairs at any time and in the most convenient manner.
- The Khajane project in Karnataka is the state government's extensive online treasury computerisation project. It has resulted in the computerisation of the state government's whole treasury related activities, and the system can track everything from the adoption of the state budget to the point of delivering accounts to the government.
- Smart Gov. (Andhra Pradesh): It has been created for application in the Andhra Secretariat to streamline processes and improve performance through workflow automation and knowledge management.

Government Initiative as Mission Mode Project (MMP)

- e-Office: Aims to improve workflow by allowing for faster search and retrieval of files and office orders, as well as digital signatures for authentication, forms, and reporting components.

- UID: The unique identification project was envisaged as an effort that would offer identification for each citizen across the country, with the primary goal of facilitating the effective delivery of social services. H would also serve as a tool for effective monitoring of numerous government programmes and plans.
- Pensions: It intends to make pension retirement related information, services, and grievance handling procedures available online to needy retirees using a combination of interactive and noninteractive components, bridging the gap between pensioners and the government.
- Banking: To streamline diverse e-services initiatives launched by separate banks.
- Post: The postal service has modernised postal services by computerising and networking all post offices with a centralised server based system, as well as establishing computerised registration centres.
- e-Governance in municipalities: The NeGP and Jawahar Lal Nehru National Urban Renewal Mission aim to improve the operational efficiency of municipal governments.
- Crime and criminal tracking system: An integrated and comprehensive system for improving the efficiency of efficient policing at all levels, particularly at the police station level, by implementing e-governance concepts and establishing a countrywide network.
- Public distribution system: A complete project that includes supply chain management, including allocation and usage, reporting, storage, and transportation of food grains.
- A platform for grievance redressal and transparency, as well as the digitisation of the beneficiary database. Shop automation at a reasonable price, for example.
- Mother and child tracking system (MCTS), Ministry of Health and Family Welfare envisions a more thorough use of ICT, supply chain management of pharmaceuticals and vaccinations, and offering ICT tools to ASHA and ANM personnel. NRHIM programme management, for example.
- e-District: Establishment of a state data centre and a common services centre via a state-wide area network.
- National land records modernisation programme: computerisation of land records project. In 1988–89, it was launched to eliminate the inherent fluxes in a manual process.

ICT for Efficiency Implementation of Welfare Scheme

- Pradhan Mantri Fasal Bima Yojna: If a farmer is required to transmit a snapshot of his destroyed crop to authorities as proof. The government will then gain access to drainage via satellite photographs of the field. Following that, the insurance claim will be deposited straight into the farmer's account. As a result, there would be no discussions about payment corruption.
- Pradhan Mantri Krishi Sinchai Yojana: Smart agriculture that measures soil moisture and then supplies water automatically through deep irrigation.
- In 2014, Prime Minister Narendra Modi introduced the 'lab to land' scheme. The goals are to increase agricultural yield while using water wisely.
- More crops per drop: Prime Minister Modi has encouraged agricultural scientists to adopt scientific methods to boost agricultural production. The local blend of traditional knowledge systems and contemporary technology is emphasised.
- Mera Gaon's Mera Gaurav: it envisions scientists picking villages, staying in touch with the selected villages, and offering technical and other pertinent information to farmers via personal visits and telephone.
- The mission for digital literacy: (Union Budget 2016–2017) To reach around 6 crore homes during the next 3 years. The ability to use digital technologies such as computers, smartphones, and the Internet is defined as digital literacy.

TRI-NETRA

- The railway board of the Ministry of Railways has begun a proposal to put TRI-NETRA

equipment on locomotives to improve the vision of locomotive pilots in bad weather.

- TRI-NETRA is an acronym that stands for Terrain imaging for diesel drivers and enhanced optical, and radar systems. It will include a high-resolution optical video camera, a high sensitivity infrared video camera, and a radar based terrain mapping system.
- These three system components will serve as the locomotive pilot's three eyes (Tri Netra). It is intended to see the landscape ahead of the moving locomotive during adverse weather by combining the photos taken by the three subsystems and creating a composite video image that will be projected on a computational monitor in front of the loco pilot.

Bharat Bill Payment System (BBPS)

- On August 8, 2014, the Reserve Bank of India (RBI) announced draught guidelines for the introduction of the Bharat Bill Payment System (BPPS). The Bharat Bill Payment System (BBPS) will operate the bill paying system in the country as a tiered structure with a single brand image, offering clients the ease of 'anytime anywhere' bill payment.
- The consolidated bill payments system will be comprised of two main sorts of entities.
    - The standard setting body will be an entity that operates the Bharat Bill Payment System (BBPS), which will cover standards about the payment, clearing, and settlement processes, among other things.
    - Bharat Bill Payment Operating Units (BBPOUs) will be the authorised operational units adhering to the BBPS criteria. The tiered structure may be strengthened further if the BBPOUs established an effective agent network.
- Participants in the BBPS will comprise approved entities such as the entity running the BBPS, BBPOUs and their agents, payment gateways, banks, billers and service providers, and other institutions as needed by the BBPS.
- The BBPS's goal is to develop an integrated bill payment system across the country and provide interoperable and accessible bill payment services to customers through a network of distributors, enabling numerous payment ways and giving rapid payment confirmation. As a result, it has been decided that the existing participants in the online commerce sector catering to the needs of bill payments as well as payment service aggregation (concerning bill payments) would be a part of BBPS.
- At the moment, the country's internet commerce space is dominated by content providers on one side and payment aggregators on the other. While the payment side may include both banks and intermediaries supporting payment service aggregation, the content side may include a diverse collection of intermediaries ranging from those hosting digital marketplaces to those facilitating merchants belonging to certain verticals. Some of these facilitator also collect and aggregate bill payments. Similarly, some content producers provide payment services through their contracts, while others rely on specialist payment aggregators.
- In the long term, the scope of BBPS could be expanded to include services that facilitate the catalogue of repetitive payments, such as school/university fees, municipal taxes/ payments, and so on, as well as other e-commerce service content as determined by the RBI from time to time.

Practice Exercise

1. Statement I: Digital India is a program to transform India into a digitally empowered society.
Statement II: Digital India is a program to institute smart cities.
(A) Both statement I and statement II are individually true and statement II is the correct explanation of statement I.
(B) Both statement I and statement II are individually true but statement II is not the correct explanation of statement I.
(C) Statement I is true but statement II is false.
(D) Statement I is false but statement II is true.

2. Consider the following statements.
The vision of digital India launched by the government of India is centred on:
I. Digital infrastructure as a utility only to senior citizens.
II. Governance and services on demand.
III. Digital empowerment of every citizen.
Which of the above statements is correct?
(A) I, II, and III
(B) I and II only
(C) I and III only
(D) II and III only

3. PRAGATI is the acronym:
(A) Proactive governance and transparency in India
(B) Proactive governance and timely implementation
(C) Primary governance for transparency in India
(D) Promoting and accomplishing governance with assured and timely implementation

4. The main goal of Government to Business (G2B) is:
I. To increase productivity by giving more access to information
II. To lower the cost of doing business
III. To create more transparency
Select the correct answer using the codes given below.
(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II, and III

5. Consider the following statement regarding Proactive Governance and Timely Implementation (PRAGATI).
I. It is aimed at addressing common men's grievances, and simultaneously monitoring and reviewing important programs and projects of GOI as well as state government.
II. It is a two-tier system.
III. Includes various technologies such as digital data management, videoconferencing, and geospatial technology.
IV. Pragati day is the fourth Friday of every month.
Select the correct code.
(A) I, III, and IV
(B) I, II, III, and IV
(C) I, II, and III
(D) I and III

6. Digital signature service has been launched on the Aadhaar platform. What is it called?
(A) Digi sign desk
(B) e-Sign desk
(C) Online sign desk
(D) None of the above

7. Consider the following objective being addressed by e-governance.
I. Helps to reduce the paperwork.
II. Provides accountability and transparency in the system.
III. Provides citizens a platform for direct participation in the governance process.
IV. Helps in elimination of hierarchy.
Select the correct answer.
(A) I and III
(B) I, II, and IV
(C) I, III, and IV
(D) I, II, III, and IV

8. The 'digilocker' launched by the government provides many advantages, which include:

I. Total protection from online thefts and hackings
II. Minimising the use of hard copy documents
III. Providing authenticity to e-documents
IV. It provides 100 MB dedicates personal storage

Select the correct answer using the codes given below.

(A) I and II
(B) I and III
(C) II and III
(D) I, II, III, and IV

9. Assertion (A): Digitise India platform provides a 'crowd sourcing' platform where each participant can particulate in digitisation.

Reason (R): Crowdsourcing provides a process of obtaining ideas, and content by soliciting contributions from a large group of people.

(A) Both Assertion (A) and Reason (R) are individually true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are individually true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

10. Which of the following statements is/are correct about the vision of digital India?

I. To prepare India for a knowledgeable future
II. Universal digital literacy
III. Portability of all entitlements through cloud

(A) I, II, and III
(B) I and II
(C) II and III
(D) Only II

11. What is/are the benefits/benefit's digitalisation of land records in India?

I. Reduction of corruption
II. Improvement in production
III. Increase in government revenue
IV. Reduction of land related dispute

(A) I, II, III, and IV
(B) I, III, and IV
(C) I, II, and IV
(D) I, II, and III

12. Which are the potential challenges in the e-governance project of the Indian government?

I. Low access to the internet
II. High smartphone penetration
III. Vulnerability to cyber attacks
IV. Social and political factors
V. Low quality mobile infrastructure

(A) I, II, III, IV, and V
(B) I, II, III, and IV
(C) I, III, IV, and V
(D) I, II, and III

13. 'e-Kranti' is related to:

(A) Agriculture production
(B) Increasing mobile connectivity
(C) Reforming government through technology
(D) Electronic delivery of services

14. Computer Emergency Response Team (CERT) is:

(A) An agency that deals with cybersecurity threats like hacking and phishing.
(B) It strengthens the security related defense of the Indian internet domain.
(C) Both (A) and (B)
(D) None of the above

15. What is an online registration system?

(A) A framework to link various hospitals across the country for Aadhar based online and appointment systems.
(B) It facilitates online appointments with the various department of a different hospital using ekYC (electronic know your consumer).
(C) Both (A) and (B)
(D) None of the above

16. The various facilities provided under ORS include?
(A) Online booking of appointment
(B) Checking lab reports
(C) Checking blood availability
(D) All of the above

17. Regarding the digital India program. Which of the following statement is/are incorrect?
I. It will help to wider the digital literacy gap between served and unserved.
II. It will help in the creation of digital infrastructure.
III. It will help in rural development.
(A) Only I
(B) II and III
(C) I and III
(D) All are correct

18. Which amongst the following is the main purpose of the DigiLocker scheme/program brought by the Government?
(A) Facility for keeping digital currency
(B) Secure transactions in net banking
(C) e-KYC verification of the people
(D) Online authentication and issuing of the documents

19. PRAGATI includes:
(A) Video conferencing
(B) Digital data management
(C) Geospatial technology
(D) All of the above

20. Goals of e-governance include:
(A) Improved efficiency within government
(B) Empowering people through technology
(C) Speed up delivery of services
(D) All the above

21. If Digilocker scheme/programme is implemented well then which amongst the following is/are bound to happen?
I. Paperless authentication and issue of some documents
II. Less need for a visit by people to Government offices for getting government documents/services
III. Less need for banks and banking services
Select the correct code.
(A) I only
(B) I and II only
(C) III only
(D) II and III only

22. The various challenges to e-governance in India include:
I. Different language
II. Low literacy
III. Digital divide
IV. Access to information, quality, timely, and speed delivery services for citizens
Select the correct option using the code given below.
(A) I and II
(B) I, II, and III
(C) I only
(D) I, II, III, and IV

23. Which of the program provides a broadband revolution in rural areas?
(A) PRAGATI
(B) NOFN
(C) e-Kranti
(D) NDLM

24. Regarding DigiLocker, sometimes seen in the news, which of the following statements is/are correct?
I. It is a digital locker system offered by the government under digital India programme.
II. It allows you to access your documents irrespective of your physical location.
Select the correct answer using the code given below.
(A) I only
(B) I only
(C) Both I and II
(D) Neither I nor II

25. Consider the following advantage of employing ICT tools in the e-governance project.
I. Speed up delivery
II. Cost reduction by reducing paperwork
III. Increases transparency
IV. Increases accountability

V. Provides improved customer service
VI. Increased access to people

Select the correct code.
(A) I, II, and III only
(B) II, III, and IV only
(C) I and II only
(D) All of the above

26. Consider the following Challenges regarding employing ICT tools in the e-governance project.
I. Digital divide
II. Cost
III. Cybersecurity
IV. Resistance to change
V. Promotion of email and bullet boards

Select the correct code.
(A) I, II, and III only
(B) II, III, and IV only
(C) I and II only
(D) All of the above

27. Consider the following statement about employing ICT tools in the e-governance project.
I. Provides efficient storing and retrieval of data
II. Provides processing information and data faster than the earlier manual systems
III. Speeding up governmental processes
IV. Increasing transparency and enforcing accountability

Select the incorrect statements.
(A) I and III only
(B) II, III, and IV only
(C) I and II only
(D) None of these

28. Which amongst the following statements is/are correct?
I. A person gives a request for digital signatures to e-sign service providers, then details are forwarded to Aadhaar based e-KYC.
II. Digital signatures are directly issued by the e-sign service providers but the e-sign certifying authority can supervise the working of e-sign service providers.

Select the correct code.
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Answer Key

1. (C)	**2.** (D)	**3.** (B)	**4.** (D)	**5.** (A)	**6.** (B)	**7.** (D)	**8.** (C)	**9.** (A)	**10.** (A)
11. (A)	**12.** (C)	**13.** (D)	**14.** (D)	**15.** (C)	**16.** (D)	**17.** (B)	**18.** (D)	**19.** (D)	**20.** (D)
21. (B)	**22.** (D)	**23.** (B)	**24.** (B)	**25.** (D)	**26.** (D)	**27.** (D)	**28.** (A)		